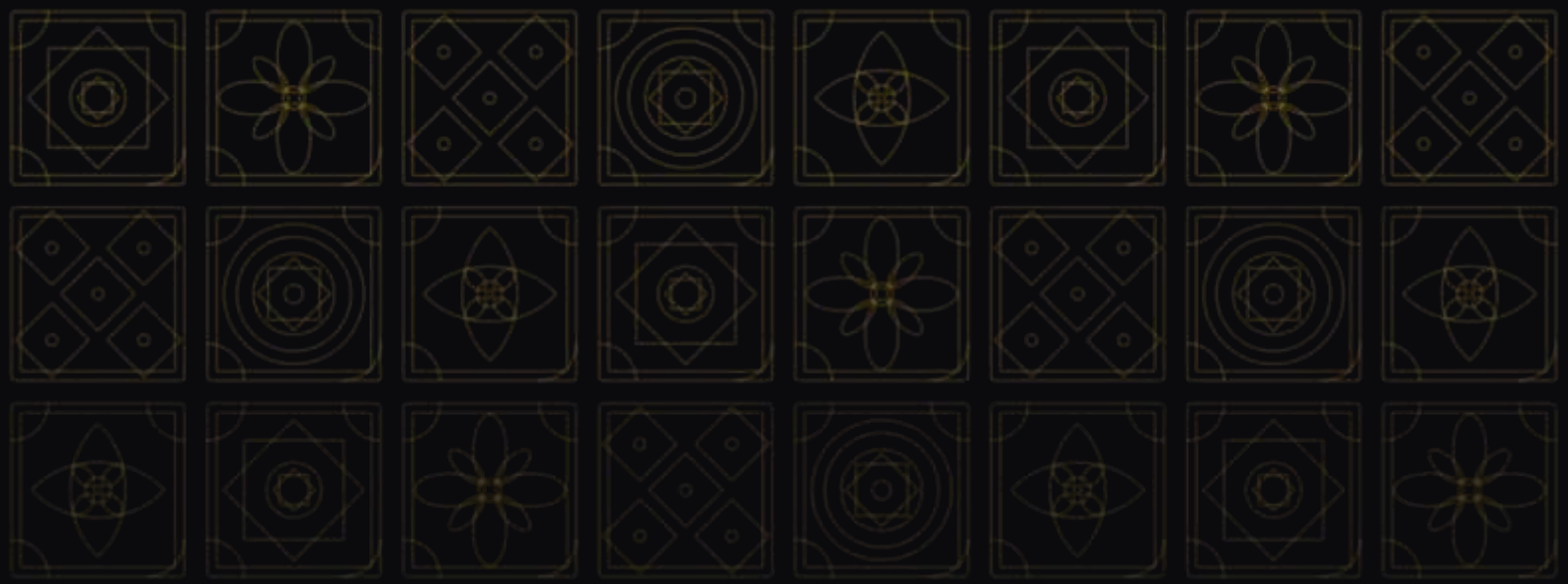




TRAINING PROGRAMME

AWS Data Engineering

Ingest, transform, govern and analyse data on AWS

27 HOURS

FOR Data analysts, developers and engineers building data pipelines on AWS

PREPARES FOR AWS Certified Data Engineer – Associate (DEA-C01), core topics



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Overview

This programme teaches participants to build data pipelines on Amazon Web Services and covers the core topics of the AWS Certified Data Engineer – Associate exam (DEA-C01). They design an S3 data lake with partitioned Parquet and Apache Iceberg tables, ingest batch and streaming data, transform it with AWS Glue and EMR, load and query Amazon Redshift Serverless, govern access and quality with Lake Formation and Glue Data Quality, orchestrate with Step Functions and Amazon MWAA, and analyse results with Athena, Amazon Managed Service for Apache Flink and QuickSight. Exam practice closes the content, and the programme ends with a final project: an end-to-end data pipeline built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Data Lakes & Ingestion	Lake, warehouse and lakehouse, S3 formats and Iceberg, batch and streaming ingestion	6 h
02	Processing & Warehousing	AWS Glue, EMR, Amazon Redshift Serverless	6 h
03	Governance, Orchestration & Analytics	Lake Formation, data quality, Step Functions, MWAA, Athena, Flink, QuickSight	6 h
04	Final Project	An end-to-end data pipeline, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	3 h
	Total		27 h

Key facts

AUDIENCE	Data analysts, developers and engineers building data pipelines on AWS
PREREQUISITES	Core AWS services (course: AWS Cloud Practitioner); Python and SQL (course: Python for Data Science), or equivalent. These are used, not taught
DURATION	27 h: 24 h of chapters + 3 h buffer
FORMAT	Short lectures followed by guided labs on a shared dataset; exam practice
LAB ENVIRONMENT	One laptop per participant with a browser and the AWS CLI; one AWS sandbox account per team, provided by the organiser, with a budget alert; MWAA and the QuickSight subscription set up by the organiser for the course, Redshift Serverless at minimum base capacity; streams, clusters and warehouses deleted at the end of each lab
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can build an orchestrated pipeline that ingests, catalogues, transforms, governs and serves data on AWS, and explain its design choices

PHASE 01

6 H

Data Lakes & Ingestion

Participants compare batch and streaming, data lakes, warehouses and lakehouses, design an S3 data lake with partitioned Parquet and Apache Iceberg tables, then ingest data with Kinesis Data Streams, Amazon Data Firehose, DMS and AppFlow, with an overview of Amazon MSK. Validated lab: a clickstream sent through Kinesis Data Streams and delivered by Firehose into date-partitioned JSON files in the raw zone of the lake.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Data Engineering Fundamentals	Batch vs streaming, data lake vs warehouse vs lakehouse	Lecture + Workshop	1.5 h
1.2	Storage & Formats	S3 data lake, partitioning, Parquet, Apache Iceberg tables (including Amazon S3 Tables)	Lecture + Lab	1.5 h
1.3	Ingestion	Kinesis Data Streams, Amazon Data Firehose, DMS, AppFlow, Amazon MSK overview	Lecture + Lab	3 h

PHASE 02

6 H

Processing & Warehousing

Participants catalogue and transform the lake with AWS Glue crawlers, the Data Catalog and ETL jobs, see when EMR is the better choice, then design, load and query a Redshift Serverless warehouse. Validated lab: a Glue job that cleans the raw data into a curated Iceberg table, loaded into Redshift Serverless and queried.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Processing	AWS Glue (ETL jobs, crawlers, Data Catalog), EMR	Lecture + Lab	3 h
2.2	Data Warehousing	Amazon Redshift Serverless design, loading and querying	Lecture + Lab	3 h

PHASE 03

6 H

Governance, Orchestration & Analytics

Participants control who sees which data with Lake Formation, check it with Glue Data Quality and trace its lineage, orchestrate the pipeline with Step Functions and Amazon MWAA, analyse it with Athena, Managed Service for Apache Flink and QuickSight, then practise DEA-C01 questions. Validated lab: an orchestrated run that stops when a data quality rule fails, and an analyst role that can query only its permitted columns in Athena.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Governance & Quality	Lake Formation permissions, Glue Data Quality, data lineage (Amazon DataZone / SageMaker Catalog)	Lecture + Lab	1.5 h
3.2	Orchestration	Step Functions, Amazon MWAA (Apache Airflow)	Lecture + Lab	1.5 h
3.3	Analytics & Streaming Analytics	Athena, Amazon Managed Service for Apache Flink, QuickSight	Lecture + Lab	1.5 h
3.4	DEA-C01 Preparation	Practice questions and review	Lecture + Workshop	1.5 h

PHASE 04

6 H

Final Project

Teams of two or three build a complete pipeline for a provided dataset, reusing the phase labs, then present it as they would to a data team lead and tear it down.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, ingest with Kinesis and S3, catalogue and transform with Glue, query with Athena	Project	3 h
4.2	Serve & Orchestrate	Load Redshift Serverless, build the QuickSight dashboard, orchestrate with Step Functions; architecture diagram	Project	1.5 h
4.3	Demo & Teardown	Live run of the pipeline, dashboard walkthrough, teardown; questions and feedback	Project	1.5 h

FINAL PROJECT

An End-to-End Data Pipeline on AWS

Data ingested with Kinesis and S3, catalogued and transformed with Glue, queried with Athena, loaded into Redshift Serverless and shown on a QuickSight dashboard, the whole run orchestrated by Step Functions and torn down after the demo.

STACK AWS: S3, Kinesis Data Streams, Amazon Data Firehose, Glue, Athena, Redshift Serverless, Lake Formation, Step Functions, QuickSight; Python, SQL

DELIVERABLE Pipeline code, architecture diagram, dashboard, live demo